

Effects of built environment and weather on bike sharing demand: a station level analysis of commercial bike sharing in Toronto

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Abstract Bike Share Toronto is Canada's second largest public bike share system. It provides a unique case study as it is one of the few bike share programs located in a relatively cold North American setting, yet operates throughout the entire year. Using year-round historical trip data, this study analyzes the factors affecting Toronto's bike share ridership. A comprehensive spatial analysis provides meaningful insights on the influences of socio-demographic attributes, land use and built environment, as well as different weather measures on bike share ridership. Empirical models also reveal significant effects of road network configuration (intersection density and spatial dispersion of stations) on bike sharing demands. The effect of bike infrastructure (bike lane, paths etc.) is also found to be crucial in increasing bike sharing demand. Temporal changes in bike share trip making behavior were also investigated using a multilevel framework. The study reveals a significant correlation between temperature, land use and bike share trip activity. The findings of the paper can be translated to guidelines with the aim of increasing bike share activity in urban centers.

Keywords Bike share Toronto · Origin–Destination model · Temporal activity

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Introduction

Over 400 cities around the world have implemented or are planning to implement a bicycle share program (Fishman et al. 2013). Like many other cities, the City of Toronto introduced its program as an active and sustainable mobility option for the citizens of as well as visitors to Toronto. The program aims to complement the existing transit system. It provides a relatively quick means of travel within the Toronto downtown core area. The bicycles are readily identifiable as belonging to the Bike Share program, thereby reducing the concern of bike theft or vandalism for the bicycle user. Bicycling, in general, has numerous benefits. Some of those benefits are: decreasing CO₂ emissions; reducing various diseases such as diabetes and obesity; reducing traffic congestion and noise pollution by providing alternatives to auto-commuting; and increasing public transit use (Shaheen et al. 2013).

The first public bicycle share system was introduced in Amsterdam, Netherlands, in the 1960s. The “White Bikes” were distributed around the city for the public’s use. The program was aborted as the bicycles were repeatedly stolen and vandalized. In 1996, Portsmouth University in England, introduced an IT based public bicycle share system (DeMaio 2009). The program’s users accessed the bicycles using electronic swipe cards. This new generation of bicycle shared systems allowed the operator to identify the customer and provided him or her with the ability to track its use. This addition to the bicycle share concept led to a significant reduction of bicycle thefts as well as vandalism. Today, we witness the evolution of “Multi-modal Systems” that provide: GPS tracking of bicycle use; real time ridership data; moveable bike stations; and system integration with public transit modes (Shaheen et al. 2013). Typically, public bicycle share systems consist of both devoted users who pay annual subscription fees as well as casual users who can pay on a month to month or per trip basis. The user can ride the bicycle for a maximum of 30 min and then return it to the nearest station (Shaheen et al. 2013). If the user was to exceed his or her allotted 30 min, extra charges may be incurred, depending on the company’s rules of use and operation.

Bike Share Toronto was established in 2011 as part of the City of Toronto’s Bike Plan (Toronto Bike Plan 2009). Currently, the bicycle mode share for the City of Toronto is estimated to be 1.2 % (StatCan 2011). However, the mode share increases substantially in Toronto’s downtown core area to 17 % (City of Toronto 2012). It is important to note that the above percentages focus on bicycling for commuter purposes and they do not include utilitarian or recreational bike trips. Thus, the overall actual bicycle mode share may differ from the above mentioned shares.

Currently, Bike Share Toronto operates a system composed of 80 stations and 1000 bicycles. The existing stations are distributed in the Toronto downtown core with plans to expand the bicycle’s network and system capacity to a total of 3000 bicycles (Toronto Bike Plan 2009). In the first 18 months since its inception, Bike Share Toronto facilitated over 1000,000 trips and attracted 4743 subscribers (City of Toronto 2012). Unlike some of the other North American public bicycle share systems, Bike Share Toronto operates 24 h per day, and is available 365 days a year.

In this paper, we attempt to examine the effect of weather, land use, socio-demographic variables, and built environment attributes on Bike Share Toronto throughout the entire year of 2013. Using real time bicycle ridership data provided by Bike Share Toronto, factors affecting trip attraction and trip generation are investigated. Further on, the shortest and most bicycle friendly routes between each station pair in the system are generated and,

through utilizing a station-to-station origin destination model, levels of service attributes are then calculated. We also briefly touch on the frequency of trips within the different hour groups of the day, focusing on the distinctive patterns of registered and casual members. This investigation is extended to include the trip frequency of users during weekdays and weekends. Finally, a multilevel modelling technique is employed to examine the trip distribution of bike share users throughout the four seasons of the year, and the different hour blocks of a day. The objective of the study is to provide a planning tool that can be used by transportation planners and city officials to predict the total trip rates of various temporal scales generated at or attracted to potential station locations and/or to assess the effect of potential bicycle infrastructure development on bicycle share ridership.

Literature review

It is no secret that land use and the built environment affect mode choice and travel behavior. Ewing and Cervero (2001) authored a review listing population and employment density and mixed land use locations as variables that lead to an increase in trip frequency. The authors cited the fact that trip frequencies are likely to increase with the availability of proper supporting infrastructure (sidewalks for pedestrians, bike lanes for cyclists...) and fewer four-way intersections. Moreover, the review states that transit, bike and walk mode shares typically increase in zones of mixed land use and high population and employment density, while that of cars usually decreases, in comparison to suburban neighborhoods. Further, many more studies focused on the effect of land use, the built environment and weather variables on cycling in particular. Research from the United States and Europe (Saelens et al. 2003, Fraser and Lock 2011) state neighborhood connectivity, safety (traffic and non-traffic related) and population density as factors driving cycling. Other research (Saneinejad et al. 2012) has investigated the effect of weather variables such as wind, precipitation and temperature on cycling. The factors, apart from temperature, were negatively correlated with biking.

As for bike share systems, research on the topic is on the rise. North American cities have been investing heavily in biking behavioral research and cyclist friendly policies since the birth of the 21st century (Pucher et al. 2011). On the forefront of such policies lie public bike share programs. Previous research showed that public bike share users tended to be less experienced than regular bike users and therefore had different characteristics than that of bicycle riders for whom bicycle use is part of their established behaviour and routines (Bachand-Marleau et al. 2012). That is, studying public bike share users' behaviour is required to fill this gap in the literature, and is likely to provide pertinent information in regards to policy development for bicycle use in urban settings.

Few of the studies used data from customer satisfaction surveys to analyze users' socio-demographic and ridership characteristics. Bachand-Marleau et al. (2012) used data on BIXI users in Montreal, Quebec, (Canada). He concluded that the location of docking stations close to the origin point of potential users can increase ridership (Bachand-Marleau et al. 2012). A recent study by Fishman et al. (2013) administered an online survey to members of both the bike share and the general community in Melbourne and Brisbane, Australia. The study investigated the relationship between socio-economic data and ridership activity with bike share membership. The conclusion stated that middle-aged males

of relatively high income constitute the primary market for bike share (Fishman et al. 2015).

Other studies attempted to understand the feasibility and potential of bicycle share systems. North American cities such as Ottawa, Vancouver, Seattle, and New York, through the help of and/or collaboration between consulting firms and government institutions, conducted several feasibility studies on public bicycle share systems. These studies hypothesized several potential station locations and determined, using intuition and previously published studies on cycling behaviour, the various factors that can promote systems' bicycle ridership. Factors such as population and employment density, bicycle infrastructure, socio-demographic characteristics, land use as well as the built environment were investigated (National Capital Commission 2009; Curran 2008, Gregerson et al. 2011, Burden and Barth 2009).

A third category of studies used ridership data from systems that are currently in operation to predict and hypothesize the number, location and distribution of stations in other cities worldwide. For instance, Krykewycz et al. (2010) used ridership data from peer European cities to estimate demand for a hypothetical bicycle share ridership system in Philadelphia, Pennsylvania (USA) (Krykewycz et al. 2010). In addition, Maurer (2011) used a pair-wise suitability analysis to understand the effects of variables such as job density, household income, and alternative commuters on public bicycle share ridership to propose the locations of bicycle stations in Sacramento, California (USA). Ridership data from the Nice Ride in Minneapolis-Saint Paul, Minnesota (USA) was used to perform the analysis. In this research the time period for the study was narrowed down to the month of August. The analysis showed a positive correlation between the number of docks per station and users' ridership. On the other hand, higher proximity to railway stations was negatively correlated with station use (Maurer 2011).

Recently, a few studies used real time bicycle ridership data to examine factors that affect public bicycle ridership demand. Gebhart and Noland (2014) used real time ridership data for Capital Bikeshare in Washington D.C. (USA) to investigate the impact of weather variables and proximity of bike share stations to metro stations on ridership levels. Reduced ridership was correlated with cold temperatures, rain, and high humidity levels (Gebhart and Noland 2014). Buck and Buehler (2012) investigated the influence of bicycle infrastructure, population density, land use mix around stations, and the number of households without a car using bicycle share systems using ridership data from Capital Bikeshare (Buck and Buehler 2012). The study was conducted using data gathered during the months of September to March. Buck and Buehler concluded that locating stations near bicycle lanes may increase ridership. In addition, Daddio (2012) performed a regression analysis on the usage of Capital Bikeshare stations during the month of October. The study concluded that proximity to retail outlets and the metro rail was positively correlated with trip generation while locating stations away from the centre of the bicycle share system tended to reduce ridership (Daddio 2012). Socio-demographic factors were also investigated and results showed that a bicycle share system was predominantly used by white, middle-aged riders.

Wang et al. (2012) conducted a similar study using data from Nice Ride, Minnesota, which only operated from the months of April to November. The study evaluated the effect of socio-demographic, land use, built environment and transportation infrastructure variables on bicycle share ridership. Station proximity to high job density and food serving enterprises was found to be correlated with higher ridership levels. Rixey (2013) explored the influence of socio-demographic characteristics such as education, income, and employment and population density on monthly ridership data from three United States

based operators (Rixey 2013). Built environment factors such as proximity to colleges and parks were included in the analysis. Rixey concluded that bike share ridership is positively correlated with the socio-demographic variables under study, and bike share activity is likely to increase as station location is selected in proximity to city attractions and other bike share stations. Nair et al. (2013) studied Paris's public bicycle share system. The study investigated system characteristics, temporal flows and intermodal use (Nair et al. 2013). Nair et al. concluded that proximity of stations to transit stops drives system use. Hampshire and Marla (2012) used real time data from the months of May to September for Barcelona's and Seville's bicycle share systems (Hampshire and Marla 2012). The ridership data was aggregated on the sub-city district (SCD) level and the analysis of ridership use was conducted on an hourly level. Population and employment density as well as land use attributes were found to be correlated with higher public bicycle use. In addition, higher ridership levels were correlated with the increase morning work-oriented trips.

Imani et al. (2014) investigated factors affecting bicycle share demand at the station level using real time ridership data. The results showed that stations close to major roads had lower trip activities compared to stations that were situated around minor roads and bicycle lanes. A number of land use and built environment variables, temporal characteristics and weather variables such as temperature were investigated. The study concluded that the installation of a greater number of stations in proximity to existing stations rather than increasing existing stations' capacities would have a greater impact on bicycle share ridership. However, that bicycle share system only operated during summer months and thus, similar to many previously mentioned studies, the study did not provide a full understanding of bicycle share ridership levels in different seasons during the year. A summary of bike share ridership literature can be found in Appendix section "[Literature review summary of related bike share studies](#)".

In this paper, data from Bike Share in Toronto, Ontario, Canada is used. Trip attraction and trip generation models for various temporal scales were developed to study factors affecting public bike share demand at the station level. The analysis was performed on Toronto's Bike Share system that operates year round and provided rich data on seasonal bike demand by investigating detailed weather variables. Nevertheless, as opposed to the current literature on the topic, this study goes one step further by analyzing the variables affecting demand along a proxy path of the bike share user by developing a station pair (origin–destination) regression model. Different factors were investigated including detailed weather variables, zonal level socio-demographic characteristics, land use, built environment and levels of service attributes. The investigation of temporal bike share activity using a multilevel framework, along with the formulation of an origin destination model, presents a unique contribution to the growing literature on bike share programs.

Study area and data description

To investigate the factors affecting public bicycle trip generation and attraction, weather information, bicycle ridership data, socio-demographic characteristics, land use and built environment attributes were amalgamated to create a complete and all-inclusive dataset.

Hourly weather data collected at the Billy Bishop Toronto city airport (also known locally as the 'Toronto Island Airport') weather station was acquired by Environment Canada. The weather station is located within 5 km of all bike share stations. The database included temperature, wind chill, humidity index (humidex), snow on ground, wind speed,

precipitation and relative humidity. Sub-zero temperature values were adjusted based on wind chill measures that combined the effect of wind speed and the measured temperature. Similarly, above zero temperature values were adjusted using humidex levels that combined the effects of humidity and the reported temperature values. Using the formulas for wind chill and humidex (Environment Canada 2014), the perceived temperature was estimated. The perceived temperature is a better indicator of the overall feel of the bike user, and therefore, the perceived temperature is used in this analysis.¹ Moreover, the temperature variables was categorized into five ranges: Very Cold (below 0 °C), Cold (between 0 and 10 °C), Cool (between 10 and 20 °C), Warm (between 20 to 30 °C) and Hot (30 °C or more). The categorization of perceived temperature help better indicates the bike share user's temperature preferences for using Bike Share Toronto.

Ridership data for the year of 2013 was made available for this study by Bike Share Toronto. The dataset included trip start time and date, trip end time and date, trip duration, start station, end station, number of docks per station and bicycle id. This allowed for a comprehensive temporal analysis based on the date and time of each trip as well as spatial analysis based on the number of trips generated to and attracted by every station. The availability of data in such a form eliminated possible errors incurred due to rebalancing operations. The data was analyzed at the station level and trips were aggregated on various temporal scales to account for the variability of weather effects. Figure 1 shows the distribution of the 80 bike share station, street network, and bike infrastructure network, while Fig. 2 shows the trip generations and attraction propensities by the stations.

After “cleaning” the data set by removing invalid trip records (e.g. bike trips of durations less than 30 s), the total number of trips completed using Bike Share Toronto in the year of 2013 was 623,649. The average number of trips generated per station on a monthly basis was 649 ranging from 60 trips to 2072 trips per station per month. Similarly, the average number of trips attracted per station was 650 with a minimum of 82 trips and a maximum of 2218 trips.

Using the start and end stations of the trips, synthetic bicycle paths were generated using Google Map's Application Programming Interface (API) (Google Maps 2014). Bicycle paths were generated such that the travelled distance was minimized and percent of bicycle infrastructure along the route was maximized. Bicycle lanes provide one of the safest means for cyclists to travel in the City of Toronto. The route selection hypothesis is supported by a study conducted by Winters et al. on bike route safety and bike user preference in two Canadian cities; Toronto, ON and Vancouver, BC. The authors concluded that bike friendly routes—route that exhibit some form of bike infrastructure—are safer, and are generally preferred by bike users (Winters et al. 2012). Bike share users were assumed to be rationale (i.e. pragmatic), in choosing the shortest and safest path to travel on. Figure 3 shows a depiction of how google maps was used to generate the synthetic routes.

Travelled distance and trip durations were determined based on the synthetic routes. The average actual trip durations between each station pair were compared against the estimated durations. Results showed consistency of station-to-station trip duration. The spatial analysis was done in ArcMap® 10.1, using an updated version of a street network

¹ An ideal indicator if how the thermal environment affects human wellbeing is the Physiologically Equivalent Temperature (PET) (Matzarakis and Amelung 2008). However, the calculation of the PET requires variables such as vapor pressure, mean radiant temperature, metabolic rate and the physical work output of the bike share user. Thus, due to data limitations, the perceived temperature described above was instead adopted.



Fig. 1 Study area

with details on bicycle infrastructure type, if any (Habib et al. 2015). The suggested bicycle paths were used along with the street network to generate the percent of bicycle path length compared to the total path length and the total number of intersections with major roads.

Socio-demographic characteristics including zonal population and employment density were also considered. The average area of a traffic analysis zone (TAZ) in the study area is 0.258 km². The data was extracted from the Transportation Tomorrow Survey (TTS) to be used in the analysis, which is a large scale household travel survey conducted in the Greater Toronto Area (Data Management Group 2011).

In addition, the effect of bicycle share stations' proximities to subway stations and university campuses on bicycle ridership was investigated. Further on, the number of neighbouring stations within the 200 meter buffer zone along with the number of docks per station was also considered as variables in the study's analysis. A data flow treatment chart is displayed in Fig. 4.

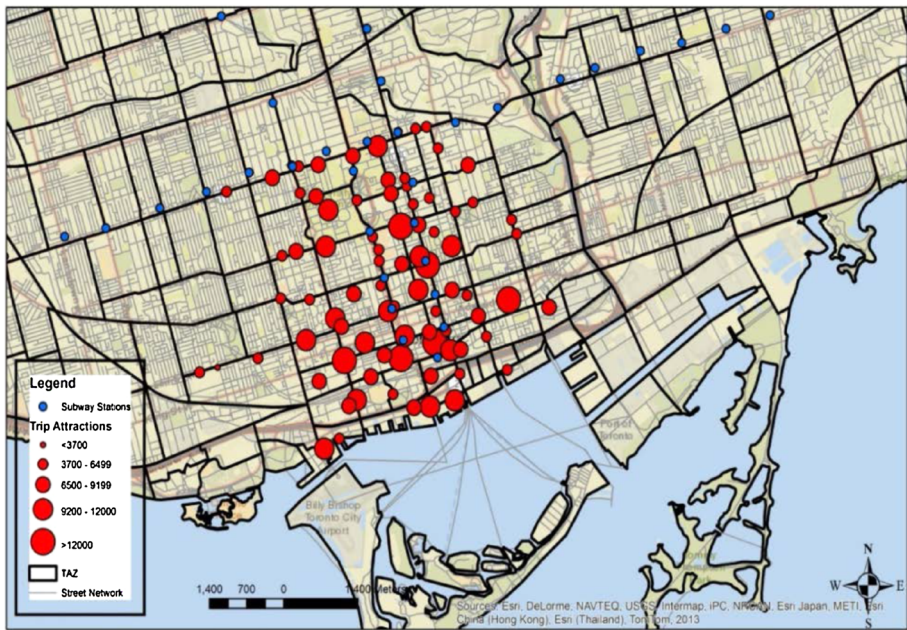
Trip time distributions

The distribution of trips throughout the different time periods of the day was investigated. A distinction was also made between registered and casual members. The day was broken down to five hour groups: Peak Morning (6:01am–10:00am); Midday (10:01am–14:00 pm); Peak Afternoon (14:01 pm–18:00 pm); Evening (18:01 pm–23:00 pm) and Overnight (23:01 pm–6:00am).

Weekday and weekend usage frequency

First, the weekday and weekend Bike Share trip frequencies per day per hour were investigated. Statutory holidays were treated as weekends. Casual members' bike share activity was consistent on both weekdays and weekends with an approximate 50/50 split. On the other hand, registered members bike share activity was more concentrated on weekdays (60.5 %) as opposed to weekends. This may be attributed to home-to-work commuters using Bike Share.

(a)



(b)

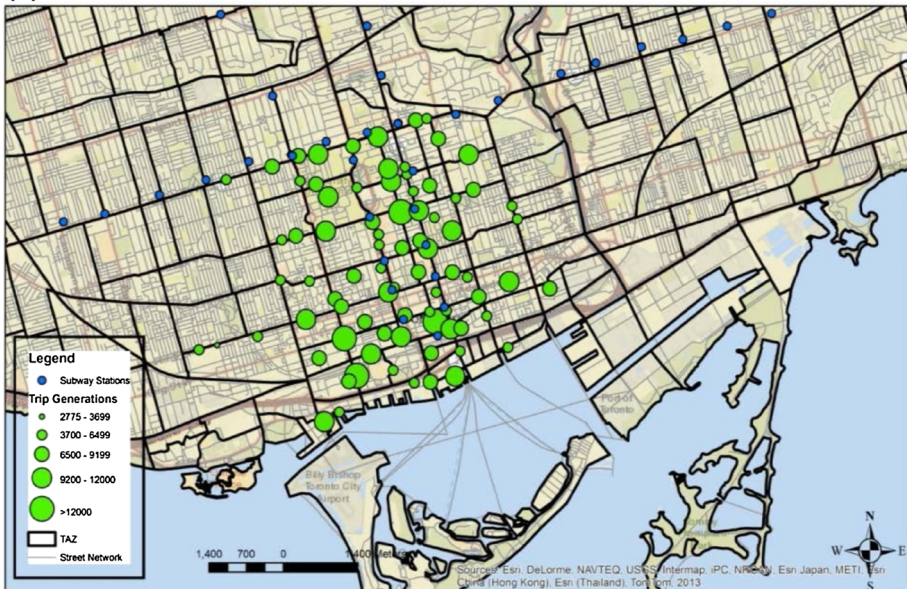


Fig. 2 **a** Trip generations at bike share stations. **b** Trip Attractions at bike share stations

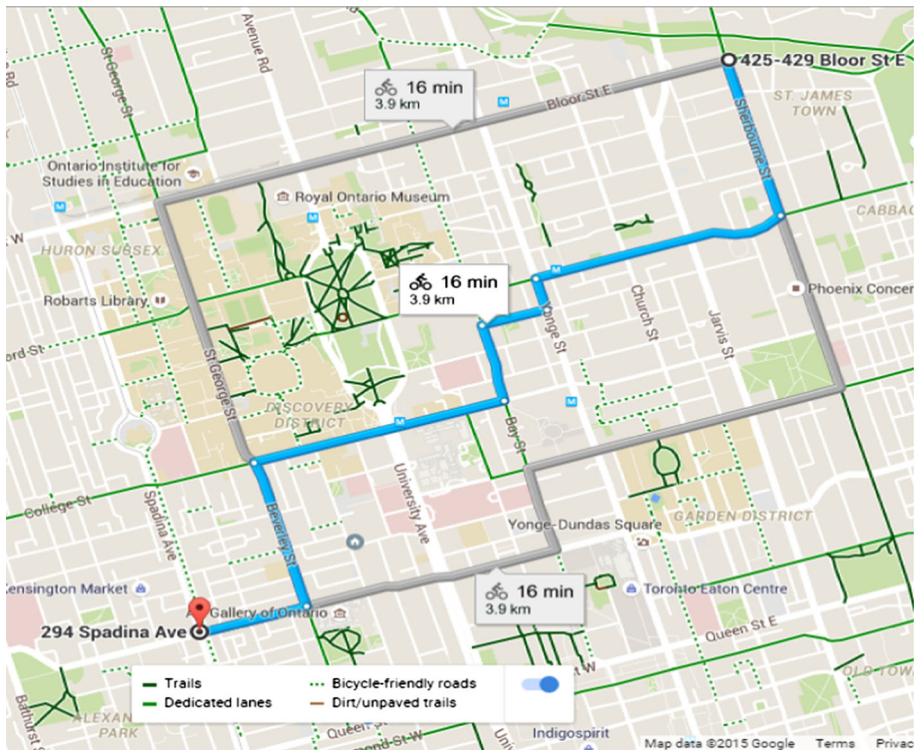


Fig. 3 Google maps O-D route depiction

Seasonal variations

Next, the year was further broken down into four seasons: fall (Sept 21–Dec 20), winter (Dec 21–March 20), spring (March 21–June 20), summer (June 21–Sept 20). In the fall season, trips were concentrated within the peak morning (24 %), peak afternoon (34 %) and midday period (22 %). As the weather got colder, the trip share of the midday period declined in the winter to 19 %. Most trips were then conducted during the peak morning (28 %) and the peak afternoon period (37 %). This may be attributed to the decline of casual use of the system while the registered members continued to use bike share to commute. Spring and summer showed similar trip distributions to the fall season with a slight increase in peak afternoon trips per hour (32 %) associated with the corresponding decrease in peak morning trips per hour (19 %).

Finally, the final dataset was prepared for econometric investigation. Table 1 shows the definitions of variables used for empirical investigations.

Methodology

In order to investigate the effects of weather, socio-economic and demographic factors, as well as land use and the built environment on bicycle share ridership, a regression analysis was performed on three different levels: trip generation, trip attraction, and station-to-

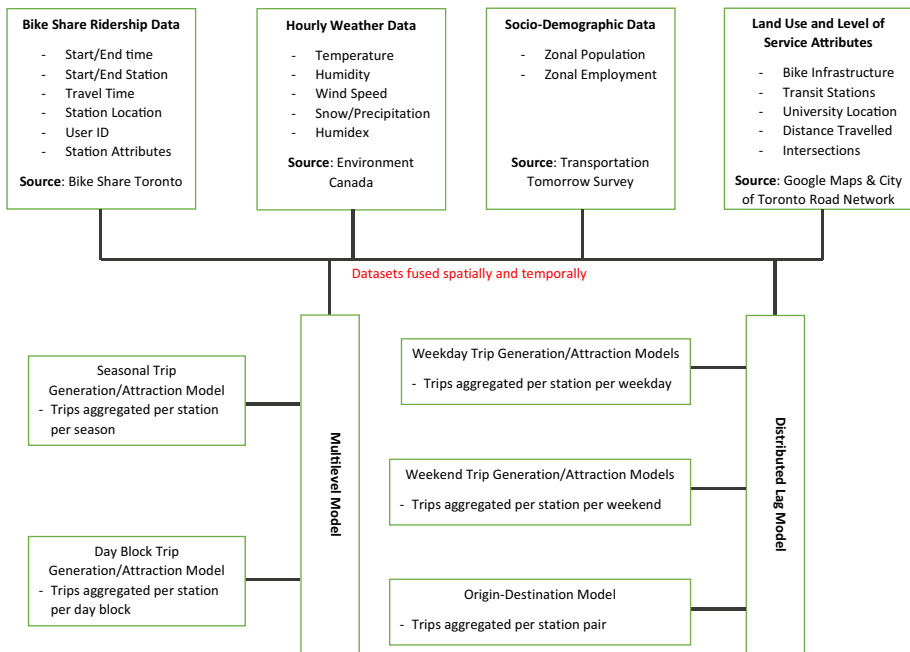


Fig. 4 Data treatment flow chart

station trips. Accordingly, five datasets were created at the station level with zonal variables where applicable. Two datasets were generated for trip attractions and trip generations on weekdays followed by another two for trip attractions and trip generations on weekends. The fifth data set was created using disaggregate trip information as such the number of trips for each origin–destination (OD) station’s pair was determined. Since the number of trips is a non-negative variable (i.e., inherently skewed from the normal distribution), the natural logarithm of the trip counts was used as the dependent variable (Table 2).

Distributed Lag model

It is reasonable to assume that responses to changes in station conditions and/or weather/built environment/socio-demographic variables may vary over time. Further, it is also reasonable to assume that in any time period, the demand for bike share users adjusts partially to its desired level (Owen and Philips 1987). These assumptions are derived from the hypothesis of serially correlated day-to-day station selection behavior of bike share users potentially indicating that bike share users, mainly registered users, can inhibit habitual behavior. To model the effect of repeated use of stations as origins or destinations throughout the week while accounting for the day-to-day trip variations, a distributed lag model was incorporated. The distributed lag model accounts for temporary changes in station status and/or weather variation. That is, unanticipated changes in system use and/or weather status may cause a temporary change in trip behavior such as station selection and/or overall bike share system use.

Table 1 Definition of variables in dataset for econometric investigation

Variable	Definition	Unit	Aggregation level
Socio-demographic			
Population density	Average population density	1000 Pers/Km ²	Zonal
Employment density	Average employment density	1000 Pers/Km ²	Zonal
Weather			
Temperature	Perceived Temperature; adjusted for wind chill and humidex. Variable categorized into five ranges with Very Cold (see Sect. 3) as base.	°C	Temporal Variable
Snow	Amount of snow on ground	cm	Temporal Variable
Humidity	Relative humidity	%	Temporal Variable
Precipitation	Amount of precipitation	mm	Temporal Variable
Built environment			
Intersections	Number of intersection with major roads between an OD pair	NA	Bike Path
Bike infrastructure	Percent of bike infrastructure compared to total bike path; bike infrastructure includes bike lanes, sharrows, park roads, contra-flow bike lanes, trails...	%	Bike Path
Docks	= 1 if the station has more than 18 bike docks; = 0 otherwise	Dummy	Station
Stations	Number of bike share stations in 200 m buffer	NA	Station
University	= 1 if the zone has university campus; = 0 otherwise	Dummy	Zonal
Transit	Number of subway/commuter rail stations	NA	Zonal
Distance travelled	Distance travelled between an OD pair	km	Bike Path

Through analyzing historical station level trip behavior, we assumed that the day-to-day operational status of a bike share station may affect future use. In other words, if the station exhibits damaged docks, inefficient balancing (always full/empty), or is relocated, such historical station shortcomings may lead to either a decline or—if the station remains in perfect condition—an increase in future use. Further on, historical variations in weather variables may temporarily affect system use in the short run.

The initial set of trip generation and trip attraction models were aggregated on a daily basis while the origin–destination station to station model was aggregated monthly.² The aggregation assisted in identifying seasonal trends and the significance of several land use parameters. In addition, certain independent variables are best captured through the implementation of daily counts such as weather variables. Moreover, the significance of a number of independent variables may vary by the type of activity conducted on weekdays

² Some origin destination (OD) pairs of bike share stations registered a very low number of daily trip counts. Therefore, OD trips were aggregated monthly.

Table 2 Summary of descriptive statistics

Variable	Unit	Min	Max	Mean	S.D.
Socio- demographic					
Population density	Per/Km ²	95	52,434	14,390	12,331
Employment density	Per/Km ²	1050	384,383	55,675	77,153
Weather					
Temperature	C	−24	42.13	14.34	12.17
Snow	mm	0	31	0.55	2.16
Humidity	%	0	100	68.21	16.33
Precipitation	mm	0	49.5	0.051	0.66
Built environment					
Transit	N/A	0	2	N/A	N/A
University	N/A	0	1	N/A	N/A
Distance travelled	km	0.147	6.958	2.464	1.202
Intersections	N/A	1	59	N/A	N/A
Bike infrastructure	%	0	100	53	28.24
Docks	N/A	7	39	N/A	N/A
Stations	N/A	0	2	N/A	N/A

and weekends. Thus, trip generation and attraction models for weekdays and weekends were generated separately.

Each distributed lag model featured a lag independent variable y_{t-1} in the form of:

$$y_t = \delta + \varphi_1 y_{t-1} + \theta_0 x_t + \varepsilon_t \quad (1)$$

The y_{t-1} variable represents the number of trips conducted on the previous day for the weekday and weekend model, and on the previous month for the station to station OD model. Moreover, the relationship between the two time periods, y_t and y_{t-1} may be denoted by the formula below:

$$y_t - y_{t-1} = \mu(y_t - y_{t-1}) \quad 0 < \mu < 1 \quad (2)$$

The μ character is a constant representing the speed of adjusting, and may be calculated by subtracting the lag coefficient from one. If μ happens to be 1, then the change is instantaneous whereas if it is 0, no adjustment takes place (Owen and Philips 1987).

Five models were generated:

1. Weekday Station-level Generation Model: The natural logarithm of the total number of trips per weekday originating from each station is the dependent variable.
2. Weekend Station-level Generation Model: The natural logarithm of the total number of trips per weekend destined to each station is the dependent variable.
3. Weekday Station-level Attraction Model: The natural logarithm of the total number of trips per weekday destined to each station is the dependent variable.
4. Weekend Station-level Attraction Model: The natural logarithm of the total number of trips per weekend destined to each station is the dependent variable.
5. Station-to-Station OD Model: The natural logarithm of the total number of trips between each OD pair per month is the dependent variable.

The five models were estimated using the Ordinary Least Square (OLS) method using the “lm” package in the statistical software “R”. A variance inflation factor analysis (VIF) was conducted on the different regression models (Appendix section “[Multivariable Regression model variance inflation factor analysis](#)”). The VIF measures the degree of multi-collinearity between the i th independent variable and all other independent variables in a linear regression model. In our investigation, all variables had a VIF of below 4 which is consistent with the rule of thumb of excessive multicollinearity found in literature (O’Brien 2007).

Multilevel/Linear Mixed Effects model: Random intercept and slope

Several previous bike share studies used multivariable linear regression to model the effect of various determinants of demand on bike ridership. However, the use of regression models does not take into account clustered data or repeated observations. Bike share user activity datasets have an inherent multilevel structure with one to many and many to one trips; trips originating from a single station and terminating at many stations. Further on, the trip time and dates may be indexed by time of day, day block (Peak Morning/Evening, Overnight, etc.), Month and/or Season. Therefore, in order to capture the temporal grouping of bike share cross-sectional data, this paper further adopts a multilevel modelling structure. The potential of modelling the variation between groups represents the feature that distinguishes a multilevel model from classical regression.

A two level model is adopted for the purpose of this study. The lower level is represented by the number of trips generated by each station. The upper level is represented by the chosen temporal aggregation. Generally, a random intercept multilevel model is written in the following form:

$$y_i = \alpha_{j[i]} + \beta x_i + \varepsilon_i, \quad i = 1, \dots, N, \quad j = 1, \dots, J \quad (3)$$

$$\alpha_j = bu_j + \eta_j, \quad j = 1, \dots, J \quad (4)$$

where y_i is an $n \times 1$ vector of random variables representing the observed value for observation i , and x_i is the $q \times 1$ vector of observed independent variable for observation i . β is an $n \times q$ matrix of regressors; it represents the coefficient for x_i . Further on, ε_i and η_j are independent error terms assumed to follow a normal distribution of mean 0 and variance σ^2 . The α_j represents a vector of varying coefficients with u_j representing predictors at the temporal level, and b is an $n \times p$ vector of second level coefficients. The u_j predictor is constant for all observations within a group while varying between the different group categories (fall, winter...). Multiple indices are used to denote individual components of a vector. For example, $\alpha_{j[i]}$ denotes the i th observation (i.e. trips) for the j th subject (i.e. temporal variable). This model is known as a varying intercept multilevel model.

A more complex form of multilevel modelling is a varying intercept and varying slope model. It can be outlined in the following form (Gelman and Hill 2006):

$$y_i = \alpha_{j[i]} + \beta_{j[i]}x_i + \varepsilon_i, \quad i = 1, \dots, N, \quad j = 1, \dots, J \quad (5)$$

$$\alpha_j = b_0u_j + \eta_{j1}, \quad j = 1, \dots, J \quad (6)$$

$$\beta_j = b_1u_j + \eta_{j2}, \quad j = 1, \dots, J \quad (7)$$

Here, α_j represents a vector of varying intercepts with u_j representing predictors at the temporal level, and b_0 is a vector of second level coefficients. In addition, β_j represents a

vector of varying coefficients with u_j representing predictors for the independent variables, and b_1 is a vector of second level coefficients. It is assumed that the random errors are also independent of the random unobserved/random effects. Both however are normally distributed.

In the case of more than 1 random variable, the model would follow the form of:

$$y_i = \alpha_{j[i]} + \beta_{1j[i]}x_i + \beta_{2j[i]}x_i + \dots + \varepsilon_i \quad (8)$$

We estimate four multilevel models:

1. Station level weekday/end trip attraction multilevel model: The natural logarithm of the total number of trips is the dependent variable. The subject level in study is the trip variation across the weekday/end day blocks.
2. Station level weekday/end trip generation multilevel model: The natural logarithm of the total number of trips is the dependent variable. The subject level in study is the trip variation across the weekday/end day blocks.
3. Station level season trip generation multilevel model: The natural logarithm of the total number of trips is the dependent variable. The subject level in study is the trip variation across the seasons.
4. Station level season trip attraction multilevel model: The natural logarithm of the total number of trips is the dependent variable. The subject level in study is the trip variation across the seasons.

The three models were estimated using the Maximum Likelihood method using the “lme4” package in the statistical software “R” (Bates et al. 2014).

Empirical results

Weekday and Weekend Distributed Lag model

Data from 80 stations was used to develop trip generation and attraction models on a weekday basis for a full year. Table 3 shows the estimation results of the two weekday models. The reported adjusted R^2 values were 0.65 and 0.63 for the trip generation and attraction models respectively. In other words, more than 65 % and 63 % of the variation in trip generation and attraction could be explained by the variables included in the models. This is a relatively high fit considering the small number of trip origins/destinations available for this analysis.

Most of the parameters were estimated with the expected signs and were statistically significant at the 95 % level of confidence. The 1 day trip activity lag was positively correlated with trip attraction and generation. This indicates that as station status (operational efficiency, docks availability) and weather conditions improve from one day to another, an increase in station use is likely to result for both trip attraction and generation purposes and vice versa. This also indicates that such occurrences can have an impact on station use. This points to the possibility that station selection is serially correlated, and that bike share users are potentially habitual in their use of the system.

Trip activities were higher in zones that had university campuses as well as transit stations, which indicated that the system users may be using bicycles to both access and egress transit stations. This is consistent with previous research done on the topic (Bac-hand-Marleau et al. 2012; Hampshire and Marla 2012). At the station level, stations with a

Table 3 Weekday trip generation and attraction models' results

Variable	Trip generation model		Trip attraction model	
	Estimate	t-value	Estimate	t-value
Intercept	0.80616	70.99572*	0.798178	69.057*
1 Day Lag	0.42741	83.95442*	0.423019	82.08158*
Humidity	−0.00285	−20.27668*	−0.002672	−18.5908*
Cold (0–10)	0.09966	22.39413*	0.09917	21.84088*
Cool (10–20)	0.18345	38.56335*	0.18348	37.86975*
Warm (20–30)	0.26184	49.48946*	0.258464	48.13447*
Hot (30 +)	0.23092	26.05083*	0.231705	25.53603*
Precipitation	−0.09840	−22.71540*	−0.09989	−22.4744*
Snow	−0.01017	−17.30261*	−0.01036	−17.8179*
Docks	0.00921	3.26503*	0.014034	16.5899*
Transit	0.02940	10.47100*	0.014371	5.018036*
University	0.05753	15.07559*	0.033203	8.51369*
Employment density	0.00036	17.78345*	0.000613	28.99715*
Population density	0.00061	5.15702*	0.000305	2.520241*
Stations	0.02147	7.23067*	0.012547	4.133362*
Adjusted R ²	0.65		0.63	
Number of observations	19,937		19,970	

* Significant at the 95 % confidence interval

higher number of bicycle docks were more likely to generate or to attract more trips. In general, the values of the parameters' estimates of the two models were close. However, the effect of the station size was found to be greater on trip attraction. The reason for that is, perhaps, Bike Share Toronto users tended to choose stations with a higher number of bicycle docks to increase their chances in finding an available dock to return their bicycles to. Otherwise they would have to travel to a nearby station that had an available dock.

The number of stations within a 200 meter buffer area was also significant and was positively correlated with the number of trip attractions and generations. Multiple stations in close proximity provide Bike Share users with multiple options to park his or her bicycle while reducing the inconvenience caused by the unavailability of vacant docking stations. Population and employment density were positively correlated with both trip attraction and trip generation. Nonetheless, the estimates for population density are more substantial for trip generation while the estimates for employment density are larger for trip attraction. This may indicate that the bike share service is being used for commuter purposes.

The results also display a positive correlation between bike share activity and temperature increase. Specifically, bike share ridership seems to be at its highest whenever the perceived temperature is between 20 to 30 degrees Celsius. Lower temperature ranges of 0 to 10 and 10 to 20 degrees are still positively correlated (in comparison to the base level of less than 0 degrees) with bike share activity. This may allude to the benefit of keeping bike share systems operational during the early fall and late spring period. Further, bike share ridership was negatively correlated with precipitation, snow on ground and humidity as such provide unfavorable weather conditions for outdoors physical activity as the risk of accidents or injury tend to be higher.

Data from 80 stations was also used to develop trip generation and attraction models on a weekend basis for a full year. Table 4 shows the estimation results of the two weekend models. The reported adjusted R^2 values were 0.64 and 0.62 for the trip generation and attraction models respectively.

The zonal population density was significant and positively correlated with both trip attraction and generation. This may be due to the increase system use of casual members on weekends. That is, casual members' trips may originate from populous zones and travel to other populous areas with greater attractions. Trip activity proved to be insignificantly correlated with zones that had transit stations. This may be explained by casual behavior again, as such members on weekends may simply use the system for leisure or utilitarian activities rather than commuting or adopting intermodal trips. The significance and correlation of the remaining variables carry a similar explanation to the weekday models. It is worth noting that variables pertaining to employment density and station proximity to universities were found to be insignificant upon estimating the models and were thus excluded. This may be explained by the nature of weekend activities.

Station to Station Origin Destination Distributed Lag Model

In order to perform the regression analysis on the number of trips from/to each origin–destination pair, an OD dataset was prepared and data from 6316 origin–destination trip assignments were used. Based on the start and end stations, bicycle routes were generated. Bicycle path measures such as distance travelled, number of intersections with major roads, and the percent of bicycle infrastructure to the total bike route were calculated. In addition, zonal attributes of the destination station zone were produced. Table 5 shows the model's results. All the estimated parameters were statistically significant and with the expected correct signs. The reported adjusted R^2 is 0.47.

Table 4 Weekend trip generation and attraction models' results

Variable	Trip generation model		Trip attraction model	
	Estimate	t-value	Estimate	t-value
Intercept	0.66662	30.48834*	0.63395	29.17529*
1 Day lag	0.41855	46.95971*	0.44981	55.90308*
Humidity	−0.00423	−14.7946*	−0.004834	−17.30249*
Cold (0–10)	0.18408	20.68284*	0.23815	29.86175*
Cool (10–20)	0.33267	34.63885*	0.37587	43.29783*
Warm (20–30)	0.40387	40.24136*	0.44503	49.23097*
Hot (30+)	0.55618	24.61798*	0.59917	26.51747*
Precipitation	−0.28295	−22.70946*	−0.27776	−21.59820*
Snow	−0.01304	−11.04181*	0.00000	−0.89580
Docks	0.02179	4.05749*	0.01335	2.43090*
Population density	0.002372	10.97116*	0.00190	8.62625*
Adjusted R^2	0.64		0.62	
Number of observations	8433		8398	

* Significant at the 95 % confidence interval

Table 5 Station-to-station origin–destination model results

Variable	Estimate	t-value
Intercept	0.439939	98.8429*
1 Month lag	0.584132	213.1208*
Docks	0.023733	9.007547*
Population density	0.000622	5.921668*
Distance travelled	−0.0421	−21.7496*
Bike infrastructure	0.005868	2.096474*
Intersections	−0.00141	−6.85681*
Transit	0.012544	5.24054*
Adjusted R ²	0.47	
Number of observations	64,287	

* Significant at the 95 % confidence interval

The results showed a negative correlation between distance travelled and the number of trip activities between each origin–destination pair. Similarly, the increase in the number of intersections between the bike path and major roads has a negative effect on the number of trips to/from each OD pair (Mahmoud et al. 2015). Major intersections inhibit heavy traffic, both automotive and pedestrian, thus bicycle users might attempt to avoid crossing such intersections due to the risk of injury, accidents or time delays in the study area. Previous research efforts obtained a significant correlation between ridership and bicycle lanes while other researchers found counterintuitive results with bicycle lanes having a negative correlation with ridership (Buck and Buehler 2012; Rixey 2013). However, their analysis focused on quantifying bike lanes in buffer zones around bicycle share stations.

In this paper, we investigated the effect of bicycle infrastructure found along users' paths from each OD pair. Results showed that the increase of bicycle infrastructure length compared to the total bicycle path significantly encourages more bicycle trips. Bicycle infrastructure and the number of intersections can be considered as a safety proxy for bicycle users.

The one month trip activity lag was positively correlated with station to station trip activity. A month long lag is a strong indicator of short term station changes such as moving a station from one zone to another. In this case, the lag variable simply states that the consistent state of a station may cause an increase of trip activity over the months and a change of station status may cause an increase or decrease of station activity. At the destination station zonal level, the population density, transit stations, and the number of docks had a positive correlation with the total number of trips, as was expected. A positive correlation with population density at the zonal level suggested that users tended to use the system for utilitarian and/or leisure purposes. The rationale behind the positive correlation with the number of docking stations and transit stations is analogous to the trip attraction and trip generation model results.

Multilevel/Linear Mixed Effects model: Random intercept and slope

A linear mixed effects model approach was adopted to study the temporal variation of bike share trip activity within the context of weather variables, socio-demographic characteristics and the built environment. Most of the parameters were estimated with the expected signs and were statistically significant at the 95 % level of confidence. The first two models (Tables 8, 9 in Appendix “Linear Mixed Effects Models' Table of Results”) investigated

the variation of trip activity along the various seasons of the year. The other two models (Tables 10, 11 in Appendix “[Linear Mixed Effects Models’ Table of Results](#)”) of trip attraction and generation investigates the variation of trip activity within the various, previously defined, hour groups of weekdays and weekends.

Seasonal trip variation and attraction was positively correlated with zones that had transit stations, zones of high population and employment density and zones of university campuses where it can be seen that the fall and winter seasons exhibited higher coefficients reflecting student use along the academic year. Bike share trip activity also showed a positive correlation with temperature, where different estimates were calculated per season.

Daily bike share trip activity was also positively correlated again with zones that have transit stations, along with population density and number of docks per station. Varying coefficients were estimated per day block. The coefficients indicate that weekday peak use of the system likely occurs during the peak morning, peak afternoon and mid-day periods whereas weekend peak use of the system likely occurs during the mid-day and peak afternoon periods.

A log likelihood ratio test was used for selecting the final models. For both the arrivals and departures model, the Chi square value obtained of each temporal scale (Seasonal and/or Day Block) was significant at a reasonable level of significance (Table 12 in Appendix “[Linear Mixed Effects Models’ Table of Results](#)”).

Conclusion and recommendations for future work

Previous studies on public bicycle share systems have been concerned with identifying the factors affecting bike share demand at the system terminals. This study builds on the current understanding of different factors driving bike share ridership. Trip attraction and trip generation models for various temporal scales were developed to study factors affecting public bike share demand at the station level. The analysis was performed on Toronto’s Bike Share system that operates year round and provided rich data on seasonal bike demand by investigating detailed weather variables. Nevertheless, this study goes one step further by analyzing the variables affecting demand along a proxy path of the bike share user by developing a station pair (origin–destination) regression model. Different factors were investigated including detailed weather variables, zonal level socio-demographic characteristics, land use, built environment and levels of service attributes.

The trip generation and trip attraction model results revealed that higher temperatures, lower humidity levels, and lower amounts of ground snow were positively correlated with bike ridership. Public bicycle share stations that were located near university campuses and transit stations experienced higher trip activities. In addition, the model’s results highlighted the importance of station capacity and station-to-station proximity in providing sufficient bicycles for trip generation and docking spaces for trip attraction.

The station pair (OD) regression model was developed based on station to station path level of service attributes along with other zonal level factors. The model’s results showed a negative correlation between distance and bicycle ridership. A positive correlation was observed between the increases of bicycle infrastructure along the path and a decrease of number of intersections with major roads. This explained the safety perception of bicycle users. On major intersections and on roads with no marked bicycle paths cyclists would be exposed to higher risks of collisions.

The seasonal and day block trip variation was investigated using a multilevel/linear mixed effect modelling technique. Temperature, station capacity, zonal population and employment density and station proximity to transit and universities were positively correlated with bike share trip activity.

The developed models can be used to predict the total monthly trip rates generated at or attracted to potential station locations. This can help bicycle share systems to examine possible locations of new stations and to choose the one that maximizes the total bicycle ridership. In addition, the station pair model can be used to estimate the distribution of those trips between new and existing stations. The model can also be used to assess potential bicycle infrastructure development based on expected bicycle routes from and to potential station locations.

In this study, suggested bike paths based on the hypothesis of utilizing bike infrastructure and following the shortest paths were used to conduct the station pair origin destination analysis. Actual users' paths can be mapped via GPS tracking units or through the utilization of mobile applications in order to enhance the developed understanding of public bike share users' preferred routes. In addition, bike share programs are implemented in major cities around the world, thus users interact with heavy traffic and congested roads. Understanding the effect of traffic, both auto and pedestrian, can assist city planners and bike share system operators in making better decisions on station location choices.

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Appendix 1: Literature review summary of related bike share studies

Author(s)	Location	Method	Time of year	Significant findings
Bachand-Marleau et al. (2012)	Montreal, Canada	Online Survey	April to November	Proximity of docking stations to residential housing increases bike share trip frequency
Gebhart and Noland (2014)	Washington D.C., USA	System Ridership Data	Year round	Reduced ridership was correlated with cold temperatures, rain, and high humidity levels
Buck and Buehler (2012)	Washington D.C., USA	System Ridership Data	September to March	Station proximity to bicycle lanes may increase ridership
Daddio (2012)	Washington D.C., USA	System Ridership Data	October	Proximity to retail outlets and the metro rail was positively correlated with trip generation. Locating stations away from the center of the bike share system tended to reduce ridership
Wang et al. (2012)	Minnesota, USA	System Ridership Data	April to November	Station proximity to high job density and food serving enterprises was found to be correlated with higher ridership levels

Author(s)	Location	Method	Time of year	Significant findings
Rixey (2013)	Washington D.C., Denver, Minnesota	System Ridership Data	Year Round Data	Population density, retail density, education level of riders, proximity of station to other bike share stations were found to be positively correlated with increase in bike share use
Nair et al. (2013)	Paris, France	System Ridership Data	March to July	Bike share ridership was positively correlated with station proximity to transit stops
Hampshire and Marla (2012)	Barcelona, Spain	System Ridership Data	May to September	Population and employment density were found to be correlated with higher public bicycle use
Imani et al. (2014)	Montreal, Canada	System Ridership Data	April to November	Station proximity to major roads was negatively correlated with ridership. Many smaller stations optimally distributed around the service area can lead to greater ridership as opposed to a few large stations

Appendix 2: Multivariable Regression model variance inflation factor analysis

See Tables 6, 7.

Table 6 Variance inflation factor analysis for weekday and weekend trip generation and attraction models

Variables	Weekday		Weekend	
	Generation	Attraction	Generation	Attraction
1 Day lag	1.8821	1.8435	1.8022	1.4617
Humidity	1.3442	1.3369	1.4666	1.3368
Cold	1.7700	1.7509	1.9986	1.5312
Cool	2.2485	2.2172	2.1994	1.7401
Warm	2.2661	2.2280	2.4814	1.9147
Hot	1.2709	1.2647	1.3326	1.2719
Precipitation	1.2031	1.2137	1.4684	1.4519
Snow on Ground	2.0089	2.8051	2.0833	1.0096
Docks	1.0322	2.9160	1.0066	1.0053
Population	1.1067	1.1148	1.0082	1.0072
Employment	1.1977	1.2245	–	–
Transit	1.2263	1.2137	–	–
School	1.1564	1.1420	–	–
Stations	1.0792	1.3512	–	–

Table 7 Variance inflation factor analysis for Origin–Destination model

Variables	Origin–Destination
1 Month lag	1.0915
Docks	1.0158
Population	1.0093
Intersections	2.5565
Transit	1.0166
Distance Travelled	2.8334
Bike Infrastructure	1.1416

Appendix 3: Linear Mixed Effects Models' Table of Results

See Tables 8, 9, 10, 11 and 12.

Table 8 Season trip generation linear mixed effects model results

	Intercept (t-statistic)	Temp (t-statistic)	University (t-statistic)	Transit (t-statistic)	Population (t-statistic)	Employment (t-statistic)
Fixed effects	3.018607 (65.51)*	0.01834 (5.03)*	0.07603 (3.07)*	0.07389 (2.033)*	0.001526 (2.18)*	0.000374 (3.07)*
Random effects						
Fall	3.00560 (65.23)*	0.01873 (5.32)*	0.07690 (3.19)*	–	–	–
Winter	2.92806 (63.54)*	0.02429 (7.04)*	0.09488 (3.93)*	–	–	–
Spring	3.05848 (66.37)*	0.01428 (4.14)*	0.06464 (2.68)*	–	–	–
Summer	3.08228 (66.89)*	0.01246 (3.61)*	0.05912 (2.45)*	–	–	–

* Significant at the 95 % confidence interval

Table 9 Season trip attraction linear mixed effects model results

	Intercept (t-statistic)	Temp (t-statistic)	University (t-statistic)	Transit (t-statistic)	Population (t-statistic)	Employment (t-statistic)
Fixed effects	3.0164 (64.99)*	0.01742 (5.08)*	0.04431 (1.73)	0.01509 (0.92)	0.001168 (1.62)	0.0006837 (5.46)*
Random effects						
Fall	2.9988 (64.61)*	0.01875 (5.46)*	0.04945 (1.93)	–	–	–
Winter	2.9256 (63.03)*	0.02429 (7.07)*	0.07092 (2.77)*	–	–	–
Spring	3.05734 (65.86)*	0.01433 (4.17)*	0.03230 (1.26)	–	–	–
Summer	3.08368 (66.44)*	0.01234 (3.59)*	0.02457 (0.96)	–	–	–

* Significant at the 95 % confidence interval

Table 10 Day blocks trip generation linear mixed effects model results

	Intercept (t-statistic)	Transit (t-statistic)	Docks (t-statistic)	Population (t-statistic)
Fixed effects	2.6146 (20.81)*	0.043812 (2.76)*	0.040535 (2.57)*	0.002124 (3.39) *
Random effects				
Day peak morning	3.05669 (24.33)*	0.03551 (2.24)*	0.04060 (2.57)*	–
Day midday	3.00586 (23.92)*	0.05999 (3.77) *	0.03071 (1.95)	–
Day peak afternoon	3.1742 (25.27)*	0.06774 (4.26)*	0.02615 (1.66)	–
Day evening	2.8499 (22.69)*	0.05754 (3.62)*	0.03295 (2.08)*	–
Day overnight	2.2572 (17.70)*	0.05229 (3.29)*	0.03973 (2.51)*	–
End peak morning	1.9818 (15.78)*	0.02961 (1.86)	0.05139 (3.25)*	–
End midday	2.6817 (21.35)*	0.03225 (2.03)*	0.04487 (2.84)*	–
End peak afternoon	2.6472 (21.07)*	0.03449 (2.17)*	0.04420 (2.8) *	–
End evening	2.3645 (18.82)*	0.03438 (2.16)*	0.04643 (2.94)*	–
End overnight	2.1267 (16.93)*	0.03428 (2.16)*	0.04831 (3.06)*	–

* Significant at the 95 % confidence interval

Table 11 Day blocks trip attraction linear mixed effects model results

	Intercept (t-statistic)	Transit (t-statistic)	Docks (t-statistic)	Population (t-statistic)
Fixed effects	2.6169 (21.11)*	0.01765 (1.11)	0.02794 (1.60)	0.00191 (2.78)*
Random effects				
Day peak morning	2.9649 (23.92)*	0.02945 (1.85)	0.04001 (2.30) *	–
Day midday	3.003 (24.23)*	0.03074 (1.93)	0.04132 (2.37) *	–
Day peak afternoon	3.2017 (25.82)*	0.03747 (2.36) *	0.04821 (2.77) *	–
Day evening	2.8954 (23.36)*	0.02709 (1.70)	0.03759 (2.15) *	–
Day overnight	2.3426 (18.90)*	0.00084 (0.53)	0.01842 (1.06)	–
End peak morning	1.9093 (15.4)*	0.00632 (0.40)	0.00340 (0.20)	–
End midday	2.6311 (21.23)*	0.01814 (1.14)	0.02843 (1.63)	–
End peak afternoon	2.6421 (21.31)*	0.01851 (1.16)	0.02881 (1.65)	–
End evening	2.3986 (19.40)*	0.01026 (0.65)	0.02036 (1.17)	–
End overnight	2.1801 (17.59)*	0.002852 (0.18)	0.01279 (0.73)	–

* Significant at the 95 % confidence interval

Table 12 AIC and Log-likelihood ratio test results

Model	N	AIC	logLik	Chisq	Pr(> Chisq)
Season—Generation					
Season_Gen.null		−273.91	139.96		
Season_Gen.model	296	−282.52	154.26	28.609	0.001441
Season—Attraction					
Season_Att.null		−253.16	129.58		
Season_Att.model	296	−266.2	146.1	33.042	0.000268
Day block—Generation					
Day_Block_Gen.null		−114.67	60.337		
Day_Block_Gen.model	740	−153.6	87.8	39.754	4.47e−9
Day Block—Attraction					
Day_Block_Att.null		−3.893	4.9465		
Day_Block_Att.model	740	−18.2	20.1	25.565	0.000183

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